Physical quantities, modern SI & unit conversions

Base vs derived quantities, the seven 2019 SI defining constants, dimensionless angles, and CGS conversion powers.

Measurement invariant n1 u1 = n2 u2

Every physical quantity Q equals a pure number n times a reference unit u: Q = n1 u1 = n2 u2. Smaller units demand proportionally larger numerical values:

$$n_2 = n_1 \left(\frac{M_1}{M_2}\right)^a \left(\frac{L_1}{L_2}\right)^b \left(\frac{T_1}{T_2}\right)^c \quad \text{for } [Q] = [M^a L^b T^c]$$

WHY / Quick check: 1 J into ergs ([M L^2 T^-2]): n2 = 1 * (10^3 g / 1 g)^1 * (10^2 cm / 1 cm)^2 * (1)^-2 = 10^3 * 10^4 = 10^7 erg!

WATCH / Common JEE trap (Q22): kilogram (kg), NOT gram (g), is the SI base unit of mass — 1 g is 10^-3 kg!

All seven SI base quantities & modern definitions

Since 2019, all seven SI base units are fixed by locking exact numerical values of universal constants (zero artifact drift!):

$$\mathrm{s}(\Delta\nu),\ \mathrm{m}(c),\ \mathrm{kg}(h),\ \mathrm{A}(e),\ \mathrm{K}(k_B),\ \mathrm{mol}(N_A),\ \mathrm{cd}(K_{\mathrm{cd}})$$

Memorise exact symbols: length [L] (m), mass [M] (kg), time [T] (s), electric current [A] or [I] (A), temperature [Theta] or [K] (K), amount [N] (mol), luminous intensity [J] (cd).

WHY / Notice: charge C = A s is DERIVED in SI even though elementary charge e fixes the ampere via e = 1.602176634e-19 A s!

Supplementary angles: named unit, NO dimensions!

Plane angle d(theta) = ds / r (unit: radian, rad) and solid angle d(Omega) = dA_perp / r^2 (unit: steradian, sr; full sphere = 4 pi sr) are ratios of lengths/areas:

$$[\theta] = \frac{[L]}{[L]} = [M^0 L^0 T^0], \qquad [\Omega] = \frac{[L^2]}{[L^2]} = [M^0 L^0 T^0]$$

WATCH / Exam multi-correct classic: "Does a dimensionless quantity ALWAYS lack a unit?" NO — plane angle (rad) and solid angle (sr) have SI units yet zero dimensions!

WHY / Conversely, any quantity WITH non-zero dimensions MUST possess a unit — "unitless" ALWAYS implies "dimensionless", never the reverse!

Six instant CGS / practical unit bridges

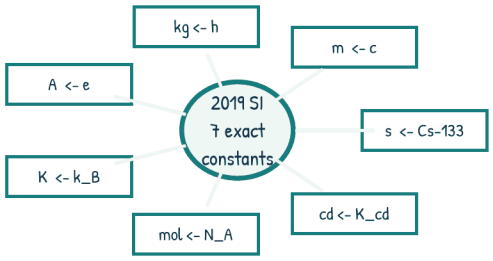
Keep standard CGS and atomic/practical conversions fluent so multi-unit numericals take 5 seconds instead of full exponent algebra:

$$1\,\mathrm{N} = 10^5\,\mathrm{dyn}, \quad 1\,\mathrm{J} = 10^7\,\mathrm{erg}, \quad 1\,\mathrm{Pa} = 10\,\mathrm{dyn/cm^2}$$

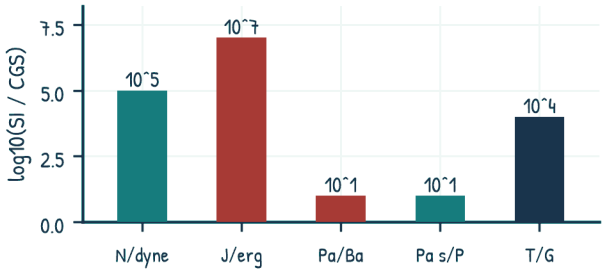
$$1\,\mathrm{Pa\,s} = 10\,\mathrm{P}, \quad 1\,\mathrm{T} = 10^4\,\mathrm{G}, \quad 1\,\mathrm{eV} = 1.60 \times 10^{-19}\,\mathrm{J}$$

WATCH / Watch viscosity trap: 1 poise = 1 g/(cm s) = 0.1 Pa s (NOT 10 Pa s!). In water at 20 C, eta ~ 1 centipoise = 10^-3 Pa s!

All seven SI base units fixed since 2019 by exact physical constants



Memorisable powers of ten when converting SI quantities into CGS units



REMEMBER / Numerical value and unit size vary inversely (n1 u1 = n2 u2); SI mass base is the kilogram (10^3 g), NEVER the gram!

Dimensional formulae I: mechanics, fluids & heat

Reconstructing every formula from its simplest defining law, twin-dimension pairs, and thermal constants.

Core mechanics, elasticity & fluid dimensions

Anchor everything on [v] = [L T^-1], [a] = [L T^-2], [F] = [M L T^-2], and [E] = [W] = [M L^2 T^-2]:

$$[G] = \frac{[F][r^2]}{[M^2]} = [M^{-1}L^{3}T^{-2}], \qquad [\gamma_{\text{surf}}] = \frac{[F]}{[L]} = [MT^{-2}]$$

$$[\eta] = [ML^{-1}T^{-1}], \qquad [Y] = [B] = [G] = [ML^{-1}T^{-2}]$$

WHY / Notice surface tension [M T^-2] matches spring constant k = F/x AND surface energy per unit area J/m^2 = [M L^2 T^-2]/[L^2]!

Rotational & quantum action quantities

Angular coordinate theta is dimensionless, so every rotational variable differs from linear mechanics purely by powers of lever arm [L]:

$$[I] = [ML^{2}], \quad [\boldsymbol{\tau}] = [ML^{2}T^{-2}], \quad [\mathbf{L}] = [h] = [ML^{2}T^{-1}]$$

WHY / Check Planck constant immediately from E = h f (P1): [h] = [M L^2 T^-2] / [T^-1] = [M L^2 T^-1] — identical to angular momentum L = m v r (Bohr quantisation L = n hbar)!

WATCH / Work and torque share [M L^2 T^-2] (P17), yet torque is NEVER measured in joules (write N m) because work = r . F while torque = r x F!

Thermal & radiation physical constants

Extract temperature [K] powers directly from P V = n R T, E = k_B T, Fourier conduction dQ/dt = -k_th A dT/dx, and Stefan-Boltzmann P = sigma A T^4:

$$[k_B] = [S] = [ML^{2}T^{-2}K^{-1}], \quad [R] = [ML^{2}T^{-2}K^{-1}\text{mol}^{-1}]$$

$$[k_{\text{th}}] = [MLT^{-3}K^{-1}], \quad [\sigma] = [MT^{-3}K^{-4}], \quad [b] = [LK]$$

WHY / Specific heat capacity c = Q/(m dT) drops mass cleanly: [c] = [L^2 T^-2 K^-1] (mass does NOT appear in specific heat or latent heat [L^2 T^-2]!).

Truly dimensionless physical ratios to recognise

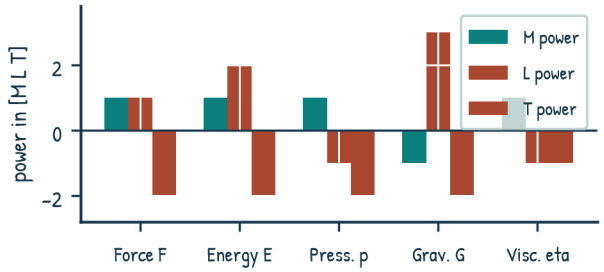
Examiners love hiding pure numbers inside multi-correct lists — spot these instantly as [M^0 L^0 T^0]:

$$\text{strain } \frac{\Delta L}{L}, \quad \text{Poisson } \nu, \quad \mu_{\text{fric}}, \quad e_{\text{restit}}, \quad n_{\text{refr}}, \quad \varepsilon_r, \quad \mu_r, \quad \text{Re}$$

WATCH / Watch specific vs relative traps: relative density (specific gravity) rho/rho_water IS dimensionless [1], whereas density is [M L^-3] and specific heat is [L^2 T^-2 K^-1]!

Also watch "velocity gradient" dv/dx = [T^-1] vs "potential gradient" dV/dx = [-E] NOT dimensionless!

Exponent signatures [M^a L^b T^c] for five foundational mechanics quantities



High-frequency exam twins: identical dimensions never imply identical physics

[M L^2 T^-2]	Work / Energy == Torque (scalar vs vector)
[M L^-1 T^-2]	Pressure == Stress == Y, B, G == Energy dens.
[M L^2 T^-1]	Angular momentum L == Planck h == Action S
[T^-1]	Frequency f == Angular vel. w == dv / dy

REMEMBER / Never memorise long exponents blindly: pick the shortest defining formula (F = G m^2/r^2, F = eta A dv/dy, P = sigma A T^4) and solve in 10 seconds!

Dimensional formulae II: electromagnetism & shortcuts

Exact [M L T A] powers of electric/magnetic fields and the golden composite blocks that bypass 4-variable algebra in JEE.

Four-step ladder from charge q = I t to V, E, R, C

Start from [q] = [A T] and potential V = W / q = [M L^2 T^-2] / [A T] = [M L^2 T^-3 A^-1]; every electric quantity falls out in one division:

$$[E] = \frac{[V]}{[L]} = [MLT^{-3}A^{-1}], \qquad [R] = \frac{[V]}{[I]} = [ML^2T^{-3}A^{-2}]$$

$$[C] = [M^{-1}L^{-2}T^4A^2], \qquad [\varepsilon_0] = [C]/[L] = [M^{-1}L^{-3}T^4A^2]$$

WHY / Notice how C = eps0 A / d gives [eps0] = [C]/[L] instantly — 5x faster and safer than squaring r inside Coulomb law!

Three-step ladder for B, magnetic flux Phi_B, L & mu0

Start from wire force F = I l B (NOT Biot-Savart!) to get [B], then Phi_B = B A and U = (1/2) L I^2:

$$[B] = [MT^{-2}A^{-1}], \qquad [\Phi_B] = [B][L^2] = [ML^2T^{-2}A^{-1}]$$

$$[L] = [ML^2T^{-2}A^{-2}], \qquad [\mu_0] = [L]/[\ell] = [MLT^{-2}A^{-2}]$$

WHY / Again solenoid inductance L = mu0 N^2 A / l gives [mu0] = [L_ind] / [L] in two seconds without touching 2 pi r!

WATCH / Watch flux quantum Phi_0 = h / (2 e): since [h]/[q] = [M L^2 T^-1]/[A T] = [M L^2 T^-2 A^-1] = [Phi_B], it has exact Weber dimensions!

Golden composite blocks (solve JEE questions in 5s!)

Nearly every JEE Main/Advanced E&M dimensional MCQ embeds known circuit/wave combinations disguised as products:

$$[RC] = \left[\frac{L}{R}\right] = [\sqrt{LC}] = [T], \qquad \left[\frac{E}{B}\right] = \left[\frac{1}{\sqrt{\mu_0\varepsilon_0}}\right] = [LT^{-1}]$$

$$\left[\sqrt{\frac{L}{C}}\right] = \left[\sqrt{\frac{\mu_0}{\varepsilon_0}}\right] = \left[\frac{h}{e^2}\right] = [R], \qquad [\varepsilon_0 E^2] = \left[\frac{B^2}{\mu_0}\right] = [p]$$

WHY / Example: "Dimensions of L I^2 / (C V)?" Spot L I^2 = energy, C V = q -> energy / q = potential [V]! Done in 4 seconds with ZERO M,L,T,A algebra!

Atomic & quantum combinations popular in JEE Adv

Three recurring combinations across Modern Physics and E&M papers:

$$\alpha = \frac{e^2}{4\pi\varepsilon_0\hbar c} \text{ (dimensionless [1]!)}, \qquad [\mu_B] = \left[\frac{e\hbar}{2m_e}\right] = [AL^2]$$

WHY / Fine-structure constant alpha ~ 1/137 has ZERO dimensions; Bohr magneton e hbar / (2 m_e) is a magnetic dipole moment m = I * Area = [A L^2]!

WATCH / Electric dipole p = q d has [A T L], whereas magnetic dipole m = I A has [A L^2]: their ratio [p / m] is [T / L] = [1 / speed]!

Shortcut composite clusters: cancel [A] immediately without expanding eps0 or mu0

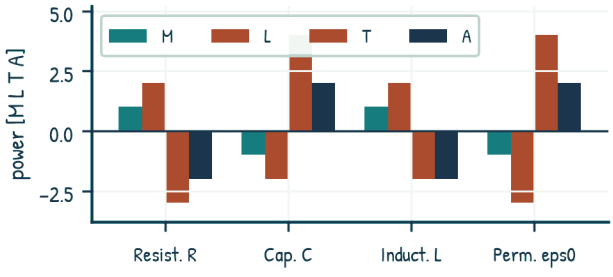
Time [T] combinations:
R C , L / R , sqrt(L C)

Speed [L T^-1] combinations:
E / B , 1 / sqrt(mu0 eps0)

Energy [M L^2 T^-2] blocks:
q V , C V^2 , L I^2 , Phi_B I

Impedance & Stress:
sqrt(mu0/eps0)=[R], B^2/mu0=[p]

Exact [M, L, T, A] exponent bars for R, C, L and vacuum permittivity eps0



REMEMBER / Never expand eps0 and mu0 in JEE MCQs when composite blocks appear: [RC] = [L/R] = [sqrt(LC)] = [T] and [1/sqrt(mu0 eps0)] = [L T^-1]!

Principle of homogeneity & transcendental functions

Matching additive terms, transcendental pure-number gates, and dimensional traps with hidden constants or logs.

Principle of homogeneity: term-by-term balance

In any physically valid equation A + B - C = D + E, every separated additive block MUST carry identical dimensions: [A] = [B] = [C] = [D] = [E].

$$\left(P+\frac{a}{V_{\text{mol}}^2}\right)(V_{\text{mol}}-b)=RT \Rightarrow [a]=[P][V_{\text{mol}}^2],\ \ [b]=[V_{\text{mol}}]$$

WHY / Homogeneity is a fast filter: s = u t + a t fails immediately because [a t] = [L T^-1] != [L] (sec. 3.4); v = u + a t^2 fails because [a t^2] = [L] != [L T^-1] (C2).

WATCH / Watch moles n! If Van der Waals is written for n moles (P + n^2 a / V^2)(V - n b) = n R T, then [a] includes mol^-2 and [b] is [L^3 mol^-1]!

Transcendental pure-number gate: exp, ln, trig

Because exp(u) = 1 + u + u^2/2! + ... adds u to 1 and u^2 to u, both the input u AND output of exp, ln, sin, cos, tan MUST be dimensionless [1]:

$$P(x,t)=\frac{\alpha}{\beta}\exp\left(-\frac{\alpha z}{k_B\theta}\right) \Rightarrow \left[\frac{\alpha z}{k_B\theta}\right]=[1] \Rightarrow [\alpha]=\frac{[k_B\theta]}{[L]}=[F]$$

Then [P] = [alpha / beta] gives [beta] = [alpha] / [P] = [M L T^-2] / [M L^-1 T^-2] = [L^2] in two lines (classic JEE Advanced question!).

WATCH / Watch integrated log forms in chemistry/thermo: ln P = C - L/(R T) hides constant C = ln P0, meaning ln(P / P0) = -L/(R T)! Standalone ln(5 metros) is meaningless!

Dimensions inside derivatives and integrals

Differentiation divides by the variable of differentiation, whereas integration multiplies by the differential dx:

$$\left[\frac{d^n y}{dx^n}\right]=\frac{[y]}{[x]^n}, \qquad \left[\int y\,dx\right]=[y][x]$$

JEE classic: find m in int dx / sqrt(2 a x - x^2) = a^m sin^-1(x/a - 1). Left side has [L] / sqrt([L^2]) = [L^0] = [1]! Since [sin^-1(...)] = [1], [a^m] = [1] => m = 0!

WATCH / Many students guess m = 1 or -1 from memorised calculus tables forgetting 2 a x - x^2 has [a] = [x], so dx / sqrt(2ax - x^2) is ALREADY dimensionless!

Necessary != sufficient: hidden constants & nth-second

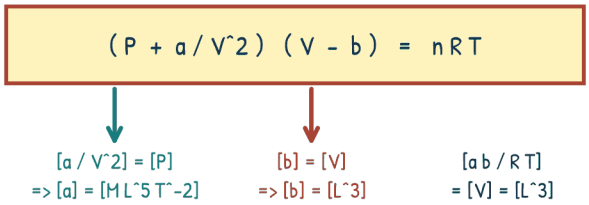
Homogeneity is NECESSARY, never SUFFICIENT: s = u t + 5 a t^2 is dimensionally consistent yet physically wrong (missing 1/2).

$$s_n=u(1\,\text{s})+\tfrac{1}{2}a(2n-1)(1\,\text{s})^2 \Rightarrow [s_n]=[L]$$

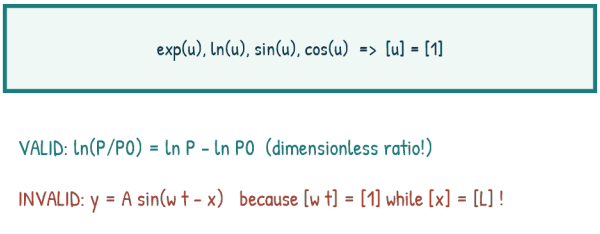
WHY / FAMOUS JEE TRAP: displacement in the nth second s_n IS dimensionally valid because hidden factor (1 second) multiplies u and a (1 s)^2 on the right!

WATCH / Hidden dimensional constants also lurk in empirical formulas like v = 331 + 0.6 T_C where "331" carries m/s and "0.6" carries m/(s K)!

Van der Waals homogeneity check: additive partners share identical dimensions



Power series u + u^2/2! + ... crashes unless argument u is pure dimensionless [1]



REMEMBER / You can ONLY add, subtract, or equate terms of identical dimensions, and arguments of exp, ln, sin, cos MUST be dimensionless [1].

Deriving relations, custom base units & limits

Solving exponent systems cleanly, changing base units to [F, V, T] or [c, G, h], and the four hard limits of dimensional analysis.

Deriving scaling laws by matching exponents

Write target Y = k X1^a X2^b X3^c, expand both sides in [M, L, T], equate powers of M, L, T, and solve the 3 linear equations:

$$\text{Drag (E2)}: \; F = k\rho r^2 v^2; \qquad \text{Blast (E5)}: \; R = k\left(\frac{Et^2}{\rho}\right)^{1/5}$$

Sensitivity check (Q21): in Sedov-Taylor R ~ E^(1/5) t^(2/5) rho^(-1/5), doubling blast energy E only raises radius R by 2^(1/5) = 1.149 (+14.9%)!

WHY / Similarly for gas sound speed v(P, rho) (Q18): v = k sqrt(P / rho) (Newton k = 1 gives 289 m/s; Laplace k = sqrt(gamma) gives 343 m/s).

Changing base units to [F, V, T] or [E, P, V] (JEE #1!)

When JEE asks for dimensions of mass or density in custom base system {Force F, Velocity V, Time T}, avoid big matrices by algebraic substitution:

$$[M] = \frac{[F]}{[a]} = \frac{[F]}{[V/T]} = [F^1 V^{-1} T^1], \qquad [\rho] = \frac{[M]}{[VT]^3} = [F^1 V^{-4} T^{-2}]$$

WHY / Test whether three proposed base quantities {X, Y, Z} are VALID independent bases using the determinant of their exponent matrix in [M, L, T]:

WATCH / CRITICAL CHECK: {Force F, Mass M, Acceleration a} CANNOT form a valid base system because F = M a makes det = 0 (cannot build [L] or [T] separately)!

Planck scales from fundamental constants (c, G, h)

Combining relativistic c [L T^-1], gravitational G [M^-1 L^3 T^-2], and quantum h [M L^2 T^-1] uniquely defines Planck length, time, and mass:

$$\ell_P = \sqrt{\frac{Gh}{c^3}}, \qquad t_P = \frac{\ell_P}{c} = \sqrt{\frac{Gh}{c^5}}, \qquad m_P = \sqrt{\frac{hc}{G}}$$

WHY / Fast exam check: once you memorize or solve l_P = sqrt(G h / c^3), dividing by c gives t_P instantly without solving a second 3x3 system!

Same trick with {e^2/(4 pi eps0), c, G}: solving [L] gives classical/gravitational charge scale in 15 seconds.

Four strict failure modes of dimensional analysis

State explicitly when dimensional derivation CANNOT complete a formula (sec. 3.5, C3, Q11):

1. Dimensionless constants: cannot find 2 pi in T = 2 pi sqrt(L/g) or 1/2 in (1/2) m v^2.

2. Dimensionless arguments: misses sin(theta) in torque r F sin(theta) and amplitude correction 1 + theta_0^2 / 16 in a large-angle pendulum.

WATCH / 3. More than 3 variables in mechanics (e.g. F depending on r, v, rho AND eta needs Reynolds function f(Re)!), OR sums of terms like s = u t + (1/2) a t^2.

Changing base units to (c, G, h): three linear equations determine x, y, z uniquely

$$[L] = [c]^x[G]^y[h]^z = [LT^{-1}]^x[M^{-1}L^3T^{-2}]^y[ML^2T^{-1}]^z$$

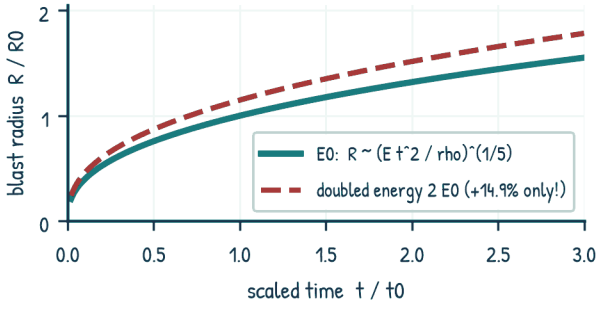
M: -y + z = 0 => z = y

T: -x - 2y - z = 0 => x = -3y

L: x + 3y + 2z = 2y = 1

=> l_P = sqrt(G h / c^3) ~ 1.6e-35 m

Sedov-Taylor shock wave: doubling explosion energy raises radius R by only 2^(1/5)



REMEMBER / In 3 base dimensions [M, L, T], dimensional analysis solves AT MOST 3 unknown powers and NEVER determines dimensionless prefactors or functions!

Significant figures, scientific notation & rounding

Exact counting rules, multiplication vs addition precision limits, IEEE round-to-even rule, and guard digits.

Five ironclad rules for counting significant figures

1. All non-zero digits and all trapped zeros between non-zeros count (1002 -> 4 sf). 2. Leading zeros NEVER count (0.00560 -> 3 sf, C4).

3. Trailing zeros AFTER a decimal point ALWAYS count (2.300 -> 4 sf; 0.00230e3 -> 3 sf, P3). 4. Unit change NEVER alters sig figs: 2.30 m = 230 cm = 0.00230 km = 2300 mm (write 2.30e3 mm!).

$$N = a \times 10^{b} \quad (1 \le |a| < 10): \quad \text{sig figs of } N \equiv \text{digits in } a$$

WATCH / Exact integers & definitions (2 in 2 pi r, 4/3 in sphere volume, speed of light c) have INFINITE significant figures and never limit an answer!

Arithmetic precision rules: +/- vs x and /

Addition & subtraction follow DECIMAL PLACES (convert all terms to the SAME power of 10 first!):

$$12.5 + 0.345 = 12.845 \to 12.8 \quad (1 \text{ decimal place, Q3})$$

Multiplication & division follow TOTAL SIGNIFICANT FIGURES (E6, C5, Q16):

$$\frac{(2.5 \times 10^{3})(4.00 \times 10^{-2})}{1.50 \times 10^{1}} = 6.667 \to 6.7 \quad (\text{limited to 2 sf by 2.5})$$

WATCH / Watch mixed powers in +/-: write 4.2e-3 + 3.15e-4 = (4.2 + 0.315)e-3 = 4.5e-3 (2 sig figs, NOT 3)!

NCERT / JEE rounding rule when dropped digit is 5

Retain 1-2 extra guard digits through intermediate steps and round ONCE at the end:

$$6.751 \to 6.8, \quad 6.750 \to 6.8 \ (\text{odd}), \quad 6.650 \to 6.6 \ (\text{even})$$

WHY / Round-half-to-even eliminates upward bias across thousands of rounded numbers when the dropped tail is an exact 5000... (if ANY non-zero follows 5, ALWAYS round up!).

WATCH / Watch subtle multi-step JEE questions like area of cube of side 7.203 m (4 sf): 6 * (7.203)^2 = 311.299 -> 311.3 m^2 (factor 6 is an exact face count!).

Matching value digits to uncertainty digits (Q20)

An uncertainty dX must match the decimal place of the last reported digit of X: never leave orphan digits beyond the error bar!

$$\text{OK}: \ 9.81 \pm 0.02; \quad \text{Bad}: \ 9.80665 \pm 0.02 \text{ or } 9.8 \pm 0.025$$

WHY / Uncertainties themselves are reported to 1 significant figure (or 2 when leading digit is 1, e.g. +-0.15 in Q17: Z = 1.00 +- 0.15).

Order of magnitude convention: write N = a * 10^b with 1 <= a < 10; if a <= sqrt(10) ~ 3.16 (or <= 5 in school texts), order is 10^b, else 10^(b+1).

Anatomy of significant digits: leading zeros locate the decimal; trailing decimal zeros count

0 . 0 0 5 6 0 x 10^3

leading zeros: NOT significant — 5, 6 ALL 3 SIGNIFICANT — trailing 0 after dot: 0 impact on count — 10^k multiplier:

Whole number 1200 m -> ambiguous! Write 1.2 x 10^3 (2 sf) or 1.200 x 10^3 (4 sf)

Two distinct rounding laws: decimal-column alignment for +/- vs total digits for x / div

ADD / SUBTRACT: fewest DECIMAL PLACES	MULTIPLY / DIVIDE: fewest TOTAL SIG FIGS
12.5 (1 dec. place)	2.5 (2 sig figs)
+ 0.345 (3 dec.)	x 4.00 / 1.50 (3 sf)
= 12.845 -> 12.8 !	= 6.667 -> 6.7 !

REMEMBER / Add/subtract keeps the FEWEST DECIMAL PLACES; multiply/divide keeps the FEWEST SIGNIFICANT FIGURES — never swap the two!

Error classification, accuracy vs precision & mean statistics

Systematic vs random vs blunders, accuracy vs precision, JEE Main mean absolute error vs sample standard deviation.

Systematic vs random vs gross blunders

1. Systematic errors shift every reading reproducibly one way: zero offset (C6), miscalibrated scale, fixed parallax angle (P6), heat loss in calorimetry. Fix by calibration/model!

2. Random errors fluctuate unpredictably (+ and -) from air draughts, vibration, eye judgement. Reduced by repeating n independent measurements.

$$\text{Accuracy} \leftrightarrow \text{low bias}; \qquad \text{Precision} \leftrightarrow \text{small scatter}$$

WATCH / Trap 3 (sec. 8): a digital watch running 5 minutes fast reading 10:04:32.18 is high-PRECISION (0.01 s resolution) yet poor-ACCURACY (5 min bias)!

NCERT / JEE Main mean-absolute-error protocol

In NCERT/JEE Main lab problems with n readings x_1, ..., x_n when standard deviation is NOT requested, compute arithmetic mean x_bar and mean absolute error:

$$\bar{x} = \frac{1}{n}\sum_{i=1}^{n} x_i, \qquad |\Delta x_i| = |x_i - \bar{x}|, \qquad \overline{\Delta x} = \frac{1}{n}\sum_{i=1}^{n} |\Delta x_i|$$

$$\delta x_{\text{rel}} = \frac{\overline{\Delta x}}{\bar{x}}, \qquad \%\ \text{error} = \frac{\overline{\Delta x}}{\bar{x}} \times 100\%$$

WHY / Note carefully: residuals (x_i - x_bar) sum to ZERO algebraically, which is why absolute bars |x_i - x_bar| (or squares below) are mandatory!

Sample standard deviation sigma vs SEM sigma/sqrt(n)

Rigorous JEE Advanced / Olympiad statistics uses Bessel (n - 1) sample standard deviation sigma for single-reading spread and sigma / sqrt(n) for the uncertainty of the MEAN (E8, P24):

$$\sigma = \sqrt{\frac{1}{n-1}\sum_{i=1}^{n}(x_i - \bar{x})^2}, \qquad \sigma_{\bar{x}} = \frac{\sigma}{\sqrt{n}}$$

Worked exemplar E8: g = {9.78, 9.82, 9.79, 9.81, 9.80} m/s^2 -> x_bar = 9.800, sum(dx^2) = 0.0010, sigma = sqrt(0.0010/4) = 0.0158, SEM = 0.0158/sqrt(5) = 0.0071 -> 9.800 +- 0.007 m/s^2!

WATCH / Trap 9: do NOT report single-shot scatter sigma (0.016) when asked for the uncertainty of the best estimate x_bar (0.007), nor confuse n - 1 with n!

Consistency check against a reference value (Q13)

Two independent measurements X +- dX and Y +- dY agree within experimental error when their difference sits inside combined uncertainty:

$$|X - Y| \le \Delta_{\text{comb}}, \qquad \Delta_{\text{comb}} = \sqrt{(\Delta X)^2 + (\Delta Y)^2}$$

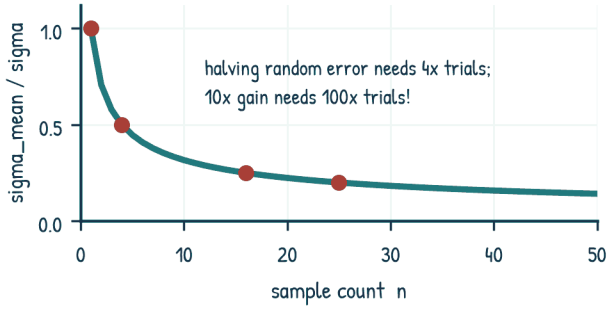
Quick check (Q13): measuring g = 9.81 +- 0.02 m/s^2 vs local reference 9.806 m/s^2 gives |9.81 - 9.806| = 0.004 <= 0.02 -> 100% consistent!

WATCH / Never delete an awkward data point ("outlier") merely to shrink sigma unless your lab log confirms a specific physical blunder!

Accuracy means unbiased mean; precision means tight cluster (small random scatter)



Standard error of the mean falls as 1/sqrt(n) — never removing systematic offset



REMEMBER / Averaging n trials cuts random standard error as 1/sqrt(n), NEVER removing systematic bias; distinguish spread sigma from mean uncertainty sigma/sqrt(n)!

Error propagation: sums, powers & coupled formulas

Absolute vs fractional rules, quadrature vs worst-case bounds, catastrophic cancellation, and the parallel-resistor trap.

Sums & differences: ABSOLUTE errors add!

For Z = A + B or Z = A - B, worst-case error adds absolute errors directly; independent random errors combine in quadrature:

$$\Delta_{\max} = \Delta A + \Delta B, \qquad \Delta_{\text{rand}} = \sqrt{(\Delta A)^2 + (\Delta B)^2}$$

Example (Q4): A = 10.0 +- 0.2, B = 4.0 +- 0.1 gives A - B = 6.0 +- 0.3 (5% error, UP from 2% and 2.5%!).

WATCH / Catastrophic cancellation (P8): for A = 1.234 +- 0.001 and B = 1.231 +- 0.001, A - B = 0.003 +- 0.002 -> relative error jumps from 0.08% to 67%!

Products, quotients & powers: log-differential rule

Take ln|Z| = p ln|A| + q ln|B| - r ln|C|, differentiate dZ/Z = p dA/A + q dB/B - r dC/C, then flip MINUS signs to PLUS for maximum uncertainty (P4):

$$\left(\frac{\Delta Z}{|Z|}\right)_{\max} = |p|\frac{\Delta A}{|A|} + |q|\frac{\Delta B}{|B|} + |r|\frac{\Delta C}{|C|}$$

Source E7 audit fix: sphere density rho = m / ((4/3) pi r^3) with m = 100 +- 1 g (1%), r = 2.00 +- 0.02 cm (1%) gives d(rho)/rho = 1% + 3(1%) = 4.0% (NOT 6% printed in E7 check box!).

WHY / For independent Gaussian random errors, quadrature replaces sum of terms by sqrt((p dA/A)^2 + (q dB/B)^2 + (r dC/C)^2).

CRITICAL JEE ADV TRAP: parallel resistors & thin lenses!

NEVER apply quotient + sum rules separately to R_eq = R1 R2 / (R1 + R2) or f = u v / (u + v), because dR1 in numerator partially cancels dR1 in denominator! Differentiate 1/R_eq = 1/R1 + 1/R2 directly:

$$-\frac{dR_{\text{eq}}}{R_{\text{eq}}^2} = -\frac{dR_1}{R_1^2} - \frac{dR_2}{R_2^2} \Rightarrow \Delta R_{\text{eq}} = R_{\text{eq}}^2\left(\frac{\Delta R_1}{R_1^2} + \frac{\Delta R_2}{R_2^2}\right)$$

$$\text{Mirror / Lens} \quad \frac{1}{f} = \frac{1}{v} \pm \frac{1}{u} \Rightarrow \Delta f = f^2\left(\frac{\Delta v}{v^2} + \frac{\Delta u}{u^2}\right)$$

WHY / Test R1 = R2 = 100 +- 1 Ohm: R_eq = 50 Ohm, exact dR_eq = 2500 * (2 / 10000) = 0.50 Ohm (1.0%), whereas naive quotient rule gives 3.0% (3x wrong!).

General single/multi-variable calculus rule

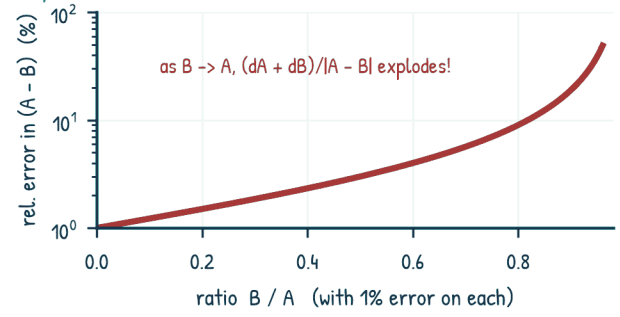
For ANY smooth function Z = f(x, y) (including logs, trig, exponentials — where power rules do not apply!), use first-order partial derivatives:

$$\Delta Z_{\max} = \left|\frac{\partial f}{\partial x}\right|\Delta x + \left|\frac{\partial f}{\partial y}\right|\Delta y$$

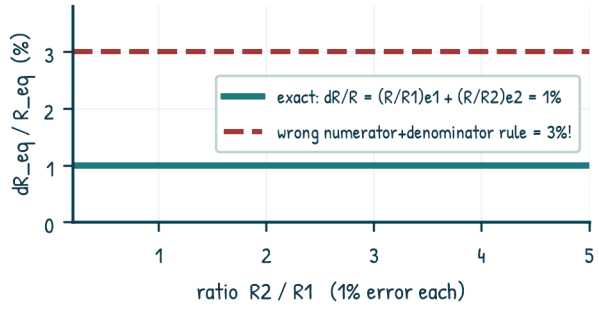
Handy specials: if Z = ln(x), dZ = dx / x (ABSOLUTE error in ln x equals RELATIVE error in x!). If Z = sin(theta), dZ = |cos(theta)| d(theta_in_RADIANS!).

WATCH / ALWAYS convert angle uncertainty d(theta) from degrees into RADIANS (multiply by pi / 180) before differentiating sin(theta) or tan(theta)!

Catastrophic cancellation: subtracting nearly equal numbers destroys relative precision



Repeating symbols trap: differentiate 1/R = 1/R1 + 1/R2 directly -> 3x smaller true bound!



REMEMBER / Sums add ABSOLUTE errors; products/powers add FRACTIONAL errors; if a variable appears TWICE (R1*R2/(R1+R2)), simplify FIRST before taking differentials!

Vernier callipers: standard, general (m/n) & retro-vernier

Exact 1 MSD - 1 VSD derivation, zero-error sign discipline, non-standard JEE Advanced verniers, and K-step coincidence.

General least-count formula (works on ALL verniers!)

Suppose n divisions on the vernier scale span m divisions on the main scale (m < n in standard forward verniers), where 1 main-scale division is 1 MSD:

$$\mathrm{LC} = 1\,\mathrm{MSD} - 1\,\mathrm{VSD} = (1 - \tfrac{m}{n})\mathrm{MSD}$$

Standard case m = n - 1 gives LC = (1/n) MSD (e.g. 10 VSD = 9 mm -> 0.1 mm; 20 VSD = 19 mm -> 0.05 mm, P5).

WATCH / JEE ADV FAVORITE: if 10 VSD = 8 MSD with 1 MSD = 1 mm, LC = (1 - 8/10) mm = 0.2 mm (NOT 0.1 mm!). Always read m AND n!

Complete reading & zero-error subtraction protocol

1. Read MSR just LEFT of vernier zero. 2. Multiply coinciding vernier mark k by LC. 3. Algebraically SUBTRACT signed zero reading e0 (E4, P15, P22):

$$x_{\text{true}} = \mathrm{MSR} + k \cdot \mathrm{LC} - e_0$$

Positive zero error (+e0, vernier 0 right of main 0, mark p coincides): e0 = +p * LC. Negative zero error (-|e0|, vernier 0 left of main 0, mark c coincides in standard n-div scale):

$$e_{0,\,\text{neg}} = -(n - c) \cdot \mathrm{LC} \quad (c = 7,\ n = 10 \Rightarrow e_0 = -3\,\mathrm{LC})$$

Worked readings & negative zero-error verification

Exemplar E4: LC = 0.01 cm, zero error e0 = -0.03 cm, MSR = 2.30 cm, 7th vernier mark aligns: x_true = 2.30 + 7(0.01) - (-0.03) = 2.40 cm!

Paper P22: LC = 0.01 cm, e0 = +0.02 cm, MSR = 2.70 cm, 5th mark aligns: x_true = 2.75 - (+0.02) = 2.73 cm.

WHY / Physical sanity check: always ask "when jaws touch, did the scale ALREADY read positive (+e0, too high -> subtract) or had not reached zero yet (-|e0|, too low -> add)?"

WATCH / Notice why negative zero coincidence uses -(n - c) * LC: if vernier 0 shifts left by tiny 0.03 cm, the 7th mark (10 - 3 = 7) lines up, NOT the 3rd mark!

Extended / retro-vernier: when 1 VSD > 1 MSD!

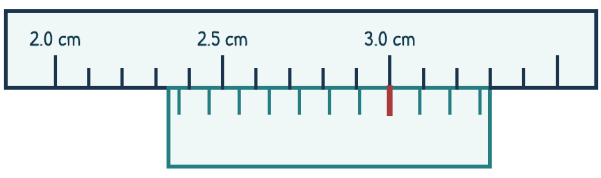
In some precision callipers (JEE Advanced 2016/2021 style), n vernier divisions span (n + 1) or (2n - 1) main-scale divisions so 1 VSD EXCEEDS 1 MSD!

$$x_{\text{raw}} = M_{\text{coinc}} \cdot \mathrm{MSD} - k \cdot \mathrm{VSD}$$

WHY / Because 1 VSD > 1 MSD, vernier gaps WIDEN to the right so coincidence steps backward (retrograde vernier has LC = 1 VSD - 1 MSD = MSD / n)!

Golden habit: writing x = M_coinc * MSD - k * VSD directly from geometry works on EVERY strange custom scale even if numbers look unfamiliar!

Drawn to exact scale: MSR = 2.3 cm + 7 x 0.01 cm = 2.37 cm (before zero offset)



7th mark aligns -> 2.30 + 7(0.01) = 2.37 cm

Closed-jaw check: algebraic rule x_true = x_raw - e0 works identically for BOTH signs

POSITIVE zero error (+e0)	NEGATIVE zero error (-|e0|)
vernier 0 RIGHT of main 0	vernier 0 LEFT of main 0
instrument OVER-READS	instrument UNDER-READS
=> SUBTRACT |e0|:	=> ADD |e0|:
x_true = x_raw - (+|e0|)	x_true = x_raw - (-|e0|)

REMEMBER / Never memorise LC = 1 mm / n blindly: ALWAYS compute 1 MSD and 1 VSD from the question stem first using LC = 1 MSD - 1 VSD!

Screw gauge, spherometer & Young modulus apparatus

Pitch and circular scale resolution, zero-offset rules, E10 source audit fix, and classic JEE practical error budgets.

Pitch, circular divisions & screw-gauge reading

If advancing the thimble through R full rotations moves the spindle linearly by D along the sleeve scale, pitch P = D / R and least count LC is:

$$\text{Pitch } P = \frac{\text{linear advance}}{\text{full rotations}}, \qquad \text{LC} = \frac{P}{N_{\text{circ}}}$$

Source E10 audit fix: pitch 0.5 mm, N = 50 -> LC = 0.01 mm. Sleeve shows 3.00 mm, circular = 22, zero error e0 = +0.04 mm -> x_true = 3.00 + 0.22 - 0.04 = 3.18 mm (NOT 3.40 mm in E10 check box!).

WATCH / Watch half-millimetre sleeve marks! With pitch = 0.5 mm, sleeve ticks show BOTH whole mm and 0.5 mm (e.g. 3.5 mm visible + 22(0.01) = 3.72 mm)!

Screw-gauge zero offset & backlash prevention

Close anvil & spindle gently using ONLY the ratchet stop: look at circular mark c on reference line with N total circular divisions:

$$\text{Below}: \; e_0 = +c\,\text{LC}; \qquad \text{Above}: \; e_0 = -(N-c)\,\text{LC}$$

Example: 100-div gauge closed reads mark 97 on reference line (zero has not arrived by 3 div!) -> e0 = -(100 - 97) * 0.01 = -0.03 mm (Q7: 5.38 - (-0.03) = 5.41 mm).

WHY / Backlash error arises from play between worn screw threads when reversing rotation direction: ALWAYS approach final reading rotating in ONE direction!

Spherometer radius of curvature formula

A three-legged spherometer (equilateral leg spacing l, screw pitch P, circular divisions N -> LC = P/N) measures sagitta height h of a spherical cap:

$$R = \frac{\ell^2}{6h} + \frac{h}{2} \approx \frac{\ell^2}{6h} \quad (h \ll \ell), \qquad \frac{\Delta R}{R} \approx \frac{2\Delta\ell}{\ell} + \frac{\Delta h}{h}$$

WHY / Geometry proof: leg circumradius is r_c = l / sqrt(3), and chord property r_c^2 = h (2 R - h) gives R = l^2/(6 h) + h/2 immediately!

WATCH / Always take ZERO reading on an optically flat glass plate first, then on the curved mirror/lens surface: h = |h_curved - h_plane|!

Searle wire, Poiseuille tube & optical bench budgets

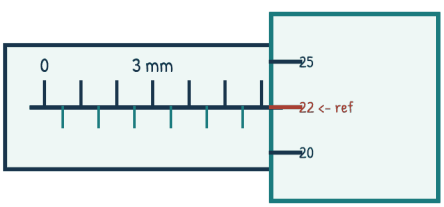
Spot which variable carries the largest exponent in JEE lab experiments to know which instrument demands highest precision:

$$Y = \frac{4MgL}{\pi d^2 \Delta L} \Rightarrow \frac{\Delta Y}{Y} = \frac{\Delta M}{M} + \frac{\Delta L}{L} + \frac{2\Delta d}{d} + \frac{\Delta(\Delta L)}{\Delta L}$$

$$\eta = \frac{\pi \Delta P r^4}{8QL} \Rightarrow \frac{\Delta\eta}{\eta} \text{ carries } 4\frac{\Delta r}{r} \; (1\% \text{ error in } r \to 4\% \text{ in } \eta!)$$

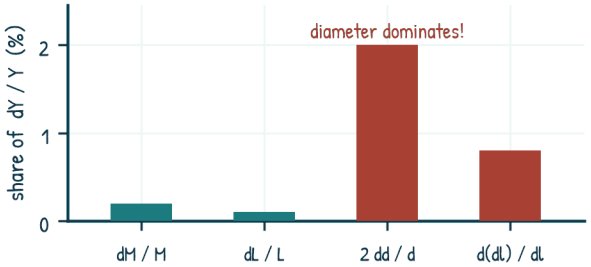
WATCH / Watch factor 2 mistake on wire diameter vs radius: d = 2 r -> dd/d = dr/r (identical percentage error — NEVER multiply dd/d by another extra 2)!

Screw gauge (pitch 0.5 mm, 50 div -> LC = 0.01 mm): corrected E10 reading = 3.18 mm



Raw = 3.00 + 22(0.01) = 3.22 mm; if e0 = +0.04 mm -> true = 3.18 mm

Searle wire Y = 4MgL / (pi d^2 dL): thin diameter d enters squared so 2 dd/d dominates



REMEMBER / LC = Pitch / N_circ; in wire/capillary formulas where diameter d enters to high powers (d^2 or d^4), measuring d with a screw gauge controls total error!

Graph linearisation, slope bounds & Fermi estimation

Straightening curved physical laws on log-log, semi-log & reciprocal axes, bounding slopes, and order-of-magnitude modelling.

Four standard linearisations every candidate must own

Choose plot axes X and Y so your physical model becomes a straight line Y = m X + c whose slope m and intercept c isolate unknown parameters cleanly:

$$y = ax^n \Rightarrow \ln y = n\ln x + \ln a \ \ (\text{slope } n,\ \text{int } \ln a)$$

$$\ln y = -bx + \ln a; \quad T^2 = \tfrac{4\pi^2}{g}L; \quad \tfrac{1}{v} = -\tfrac{1}{u} + \tfrac{1}{f}$$

WATCH / Multi-correct trap (P19): on a log-log plot of y = a x^n, y-axis intercept at ln x = 0 is ln(a) (or log10 a), NEVER raw constant a itself!

Graphical slope & worst-case bounding lines

When estimating slope quickly by hand on graph paper: (1) pick two distant points ON THE BEST-FIT LINE (NOT raw data points!); (2) draw steepest m_max and shallowest m_min lines spanning the error bars:

$$m_{\text{best}} \approx \tfrac{1}{2}(m_{\max} + m_{\min}), \qquad \Delta m \approx \tfrac{1}{2}(m_{\max} - m_{\min})$$

WHY / Non-zero y-intercept on T^2 vs L reveals systematic length offset L_eff = L_measured + r_bob: the SLOPE 4 pi^2 / g STILL gives unbiased g unaffected by constant offset r_bob!

WATCH / Taking ln(y) changes error bar height from dy to dy / y: tiny y points acquire huge log error bars and must NOT outweigh large-y points!

Fermi order-of-magnitude estimation protocol

Decompose any unfamiliar quantity into a product of 3-5 human-scale factors (random over/underestimates partially cancel in multiplication!):

$$\text{Geometric midpoint between } a \text{ and } b\text{:} \quad x_{\text{est}} = \sqrt{ab}$$

Classic atmospheric mass (P25): M_atm = P0 * (4 pi R_E^2) / g = 1e5 * 4 pi (6.4e6)^2 / 9.8 = 5.3e18 kg!
Lifetime breaths (E9): 12/min * 5.26e5 min/yr * 70 yr ~ 4.4e8!

WHY / Between bounds 10^2 and 10^4, arithmetic average is 5050 (wrong scale!) whereas geometric midpoint sqrt(10^2 * 10^4) = 10^3 hits dead center on log scale!

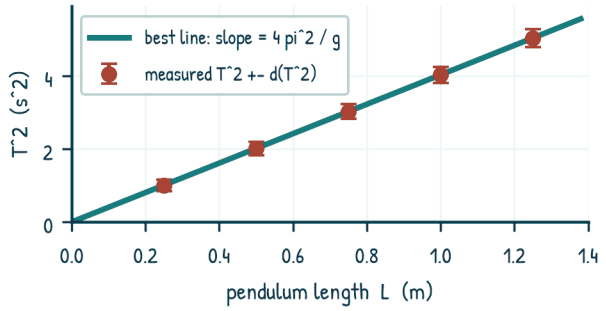
Oleic-acid monolayer thickness & Avogadro check (OL7, P33)

Dilute 1 cm^3 oleic acid into alcohol (say 1 : 400 concentration), drop volume V_oil onto water dusted with lycopodium powder to form circular film radius R:

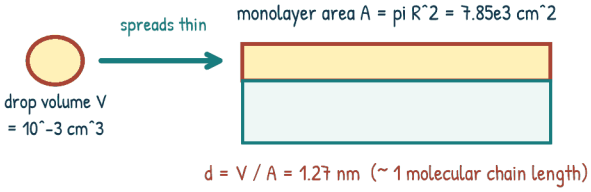
$$d = \frac{V_{\text{oil}}}{\pi R^2} = \frac{10^{-9}\,\text{m}^3}{\pi(0.50\,\text{m})^2} = 1.27\,\text{nm}$$

WHY / Gives direct tabletop proof that organic molecules have ~1 nm length! Warning (P33): estimating N_A = (M/rho)/(d * sigma) additionally assumes molecular cross-section sigma!

Linearising T = 2 pi sqrt(L/g) as T^2 vs L converts curve-fitting into one slope



Oleic-acid monolayer experiment: macroscopic volume & diameter reveal nanometre size



REMEMBER / Linearise non-linear laws BEFORE reading constants from data; in Fermi estimates, break unknown scale into 3-4 familiar factors whose logs cancel!

Master triage, multi-swing timing & top traps

Single-page exam playbook, why timing N swings beats averaging N single swings, and the ten deadliest JEE error traps.

CRITICAL LAB INSIGHT: 1/N vs 1/sqrt(N) timing law!

Suppose human start/stop reaction error on a stopwatch is dt_sw = 0.20 s and pendulum period is T = 2.00 s:

$$N \text{ continuous swings}: \; t = NT \Rightarrow \Delta T = \frac{\Delta t_{sw}}{N} \propto \frac{1}{N}$$

$$\text{Average } N \text{ separate } 1\text{–swing runs}: \; \Delta \bar{T} = \frac{\Delta t_{sw}}{\sqrt{N}} \propto \frac{1}{\sqrt{N}}$$

WHY / At N = 25 swings, continuous timing gives dT = 0.20/25 = 0.008 s (5x BETTER than 25 separate 1-swing runs giving 0.20/5 = 0.040 s in the exact same time)!

Top dimensional & sig-fig exam traps

WATCH / Angle trap: rad and sr HAVE units yet NO dimensions [M^0 L^0 T^0]; never say "unitless <=> dimensionless" works both ways!

WATCH / Hidden constants: s_n = u + (a/2)(2n - 1) IS dimensionally valid because hidden factor (1 s) sits inside every term!

WATCH / Linear dependency: before rewriting Y in custom bases {X1, X2, X3}, verify det != 0 (e.g. {Energy, Work, Time} or {F, m, a} is INVALID)!

WATCH / Decimal vs total digits: 15.4 + 0.082 = 15.5 (1 decimal place, 3 sf), whereas 15.4 * 0.082 = 1.3 (2 total sig figs from 0.082)!

Top error-propagation & instrument traps

WATCH / Sum percent trap: NEVER add percentage errors on A + B or A - B! Compute dA + dB FIRST, then divide by |A +- B|!

WATCH / Repeated variable trap: NEVER expand d(R1 R2 / (R1 + R2)) as dR1/R1 + dR2/R2 + (dR1+dR2)/(R1+R2)! Always write 1/R = 1/R1 + 1/R2 -> dR/R^2 = dR1/R1^2 + dR2/R2^2!

WATCH / Radian rule: in y = sin(theta), dy = |cos(theta)| d(theta) REQUIRES d(theta) in RADIANS (1 deg = pi/180 rad = 0.01745 rad, NEVER plug "1"!)!

WATCH / Vernier reading trap: check 1 MSD AND how many MSD equal n VSD before touching LC — custom scales like 10 VSD = 8 mm appear constantly in JEE Advanced!

Pre-submission 5-check scorecard

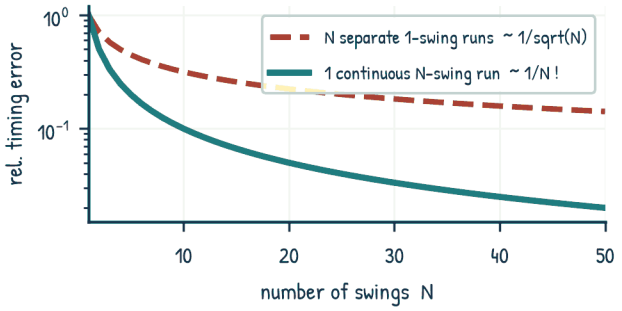
1. Did every exponent come with |p_i| PLUS signs in worst-case propagation (even for denominators C^-r)?
2. Did you check wire DIAMETER vs RADIUS (dd/d == dr/r, so d^2 contributes 2 dd/d, NOT 4)?
3. Did you subtract zero error WITH ITS SIGN: True = Raw - (+e0) and True = Raw - (-|e0|)?
4. Did you distinguish single-trial scatter sigma from standard error of the mean sigma / sqrt(n), and round uncertainty to 1 sig fig?

$$Z = (9.82 \pm 0.07)\ \mathrm{m/s^2} \quad (\text{matched decimals} + \text{SI unit})$$

Five core JEE error rules side by side — pick structure BEFORE differentiating

Sum / Diff	Power	Coupled	Instrument reading	Sample of n trials
Z = A +- B dZ = dA + dB (abs!)	Z^p B^q / C^r dZ/Z = sum |p_i| dX/X	1/R=1/R1+1/R2 dR/R^2 = sum dRi/Ri^2	True = MSR + k*LC - e0	Mean +- s / sqrt(n)

Crucial lab insight: timing N consecutive swings divides reaction error by FULL N!



REMEMBER / Timing N consecutive swings cuts stopwatch reaction error as 1/N (NOT 1/sqrt(N))! Always simplify algebraic duplicates BEFORE differentiating!

Buckingham Pi theorem & dimensional matrix rank

Rigorous linear-algebra proof via nullspace rank, capillary-gravity two-group dispersion, and atomic Bohr scales.

Exact statement via rank of dimensional matrix D

List all n physical quantities Q_1, ..., Q_n entering a law f(Q_1, ..., Q_n) = 0, and form the k x n exponent matrix D_ij across k base dimensions:

$$p = \dim\ker(D) = n - \operatorname{rank}(D) \Rightarrow F(\Pi_1, \Pi_2, \ \dots\ , \Pi_p) = 0$$

WHY / Rank trap proof: in static Hookean springs {F, k_spr, x}, symbols M, L, T appear (k = 3!), YET D columns are [1,1,-2], [1,0,-2], [0,1,0] with rank r = 2 -> p = 3 - 2 = 1 group Pi = F / (k_spr x)! Blind n - k = 0 would FAIL!

Two-Pi masterclass: capillary-gravity water waves

Phase speed v on deep water depends on wavenumber k, gravity g, density rho, and surface tension gamma: n = 5 variables across rank r = 3 [M, L, T] -> p = 2 qroups!

$$\Pi_1 = \frac{v}{\sqrt{g/k}}, \qquad \Pi_2 = \frac{\gamma k^2}{\rho g}\ (\mathrm{Bond}^{-1}) \Rightarrow v = \sqrt{\frac{g}{k}}\,\Phi\!\left(\frac{\gamma k^2}{\rho g}\right)$$

Exact hydrodynamics supplies Phi(u) = sqrt(1 + u), giving v = sqrt(g/k + gamma k / rho) with minimum speed v_min = (4 g gamma / rho)^(1/4) = 23.1 cm/s on water!

WHY / Without fluid PDEs, Pi theorem ALONE proves minimum speed scales as (g gamma / rho)^(1/4) at wavelength ~ sqrt(gamma / rho g)!

Hydrogen Bohr radius & Rydberg energy (OL8, P34)

Non-relativistic quantum electrostatics has 4 independent parameters {m_e, e, eps0, hbar} spanning all 4 dimensions [M, L, T, A] (rank r = 4):

$$a_0 = \frac{4\pi\varepsilon_0\hbar^2}{m_e e^2} = 0.0529\,\mathrm{nm}, \qquad |E_1| = \frac{m_e e^4}{32\pi^2\varepsilon_0^2\hbar^2} = 13.6\,\mathrm{eV}$$

WHY / Notice: adding speed of light c introduces n = 5th variable at rank 4 -> p = 1 dimensionless group alpha = e^2/(4 pi eps0 hbar c) ~ 1/137 (fine-structure splitting ~ alpha^2 |E_1|)!

Trinity nuclear test & prefactor caution (OL9, P28)

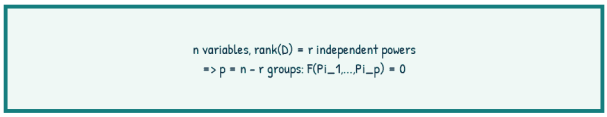
For Trinity 1945 (E = 84 TJ, t = 0.025 s, R = 100 m), naive inversion rho ~ E t^2 / R^5 (setting k = 1 in R = k (E t^2 / rho)^(1/5)) gives:

$$\frac{Et^2}{R^5} = \frac{(84\times10^{12})(0.025)^2}{100^5} = 5.25\,\mathrm{kg/m^3}\ \ (\mathrm{air:}\ \ 1.25)$$

WHY / Why off by 4.2x? Because k enters raised to the FIFTH POWER: G. I. Taylor solved the adiabatic shock ODE for air (gamma = 1.4) to get k = 1.033 (1.033^5 = 1.18; remaining factor is yield calibration)!

WATCH / Whenever solving backwards for a quantity inside a 1/5 power, a tiny 20% uncertainty in k or R explodes by factor (1.2)^5 = 2.5x!

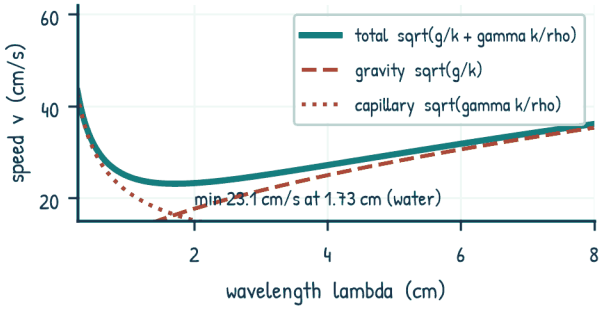
Buckingham Pi theorem via rank of the exponent matrix D (nullspace dimension n - r)



WATCH: r = rank(D), NOT blind symbol count (e.g. M T^-2 locked)!

Bohr (E, m_e, e, eps0, hbar): n=5, r=4 => p=1 => E ~ m_e e^4/(eps0^2 hbar^2)

Two dimensionless groups (n=5, r=3 -> p=2): crossover at capillary length 2 pi sqrt(gamma/rho g)



REMEMBER / Number of independent dimensionless groups is ALWAYS p = n - rank(D), NEVER blind n - k when two base dimensions only enter locked together!

Dynamical similarity, scale models & allometry

Matching dimensionless groups in wind tunnels & towing tanks, conflict of simultaneous Re + Fr matching, and Galileo scaling.

Four master dimensionless numbers of continuum physics

Divide competing force terms in Navier-Stokes / elasticity to extract universal regime numbers without solving the PDE:

$$\mathrm{Re} = \frac{\rho v L}{\eta}\ \left(\frac{\text{inertial}}{\text{viscous}}\right), \qquad \mathrm{Fr} = \frac{v}{\sqrt{gL}}\ \left(\sqrt{\frac{\text{inertial}}{\text{gravity}}}\right)$$

$$\mathrm{Ma} = \frac{v}{c_s}\ \left(\frac{\text{flow}}{\text{sound}}\right), \qquad \mathrm{We} = \frac{\rho v^2 L}{\gamma}\ \left(\frac{\text{inertial}}{\text{capillary}}\right)$$

WHY / Sphere drag F = (1/2) C_D(Re) rho pi r^2 v^2 unifies BOTH regimes: for Re << 1, C_D = 24/Re recovers linear Stokes drag 6 pi eta r v; for Re ~ 10^4, C_D ~ 0.47 gives quadratic drag!

Wind-tunnel & ship-hull model scaling dilemma

Test a 1:25 scale aircraft wing (L_m / L_p = 1/25) or ship hull in the laboratory by keeping Pi_model = Pi_prototype:

$$\text{Re match}: \ v_m = 25 v_p \ \text{ or pressurise } \ \rho_m = 25\rho_p$$

$$\text{Ship waves (Froude match)}: \ \frac{v_m}{\sqrt{gL_m}} = \frac{v_p}{\sqrt{gL_p}} \Rightarrow v_m = \frac{v_p}{5}$$

WHY / Classic IPhO trap: in the SAME fluid at fixed g, Re requires v_m ~ L^-1 while Fr requires v_m ~ L^+1/2 -> BOTH can match ONLY in a fluid of kinematic viscosity nu_m = nu_p (L_m/L_p)^(3/2)!

Mechanical & gravitational self-similarity

Scale every linear dimension of a structure or planet uniformly by factor lambda (L -> lambda L) at fixed material density rho and strength sigma_max:

$$M \propto \lambda^3, \qquad F_{\text{area}} \propto \lambda^2, \qquad \sigma_{\text{own weight}} = \frac{Mg}{A_{\text{bone}}} \propto \frac{\lambda^3}{\lambda^2} = \lambda^1$$

WHY / Galileo square-cube law: doubling an animal height without thickening proportions doubles bone stress! Constant safety factor requires bone diameter d ~ L^(3/2)!

Self-gravitating homogeneous sphere: surface g = (4/3) pi G rho R ~ R^1, free-fall / orbital period T_orb = sqrt(3 pi / G rho) ~ R^0 (INDEPENDENT of planet size R)!

Jumping animals, cooling rates & heat loss scaling

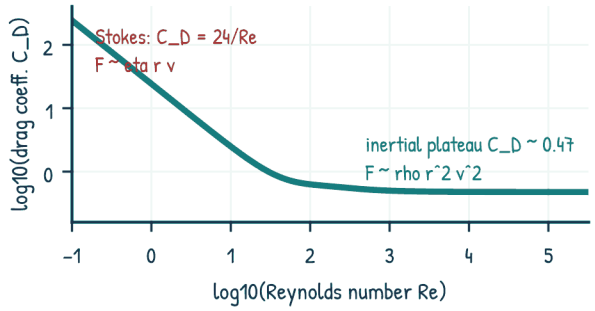
Compare maximal vertical jump height h of geometrically similar animals (muscle work W_muscle = sigma_muscle * A_muscle * s_stroke ~ L^2 * L = L^3!):

$$mgh \sim W_{\text{muscle}} \Rightarrow (\lambda^3) gh \sim \lambda^3 \Rightarrow h \propto \lambda^0$$

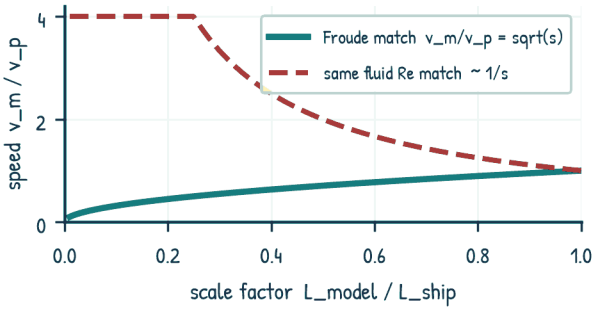
WHY / Heat-loss timescale of a warm body of heat capacity M c ~ L^3 losing heat across skin area A ~ L^2 scales as tau_cool ~ L^1: small shrews must eat continuously!

WATCH / Always state which material parameters stay constant during scaling (e.g. constant density rho vs constant total mass M changes every exponent)!

Sphere drag across six decades of Re = rho v d / eta bridges Stokes to quadratic drag



Why ship tanks cannot match Froude AND Reynolds simultaneously in water



REMEMBER / A scale model reproduces full non-linear physics faithfully IF AND ONLY IF every relevant dimensionless Pi group has the exact same numerical value!

Least-squares regression, normal equations & errors

Exact derivation of OLS slope/intercept, standard errors sigma_b and sigma_a, x-vs-y asymmetry, and weighted fits.

Normal equations & centroid theorem (OL1, P18)

For n pairs (x_i, y_i) with exact x_i and independent equal-variance vertical errors sigma_y, minimise S(a, b) = sum_i (y_i - a - b x_i)^2:

$$b = \frac{S_{xy}}{S_{xx}} = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sum (x_i - \bar{x})^2}, \qquad a = \bar{y} - b\bar{x}$$

WHY / Setting dS/da = 0 immediately forces y_bar = a + b x_bar: EVERY unconstrained best-fit line passes strictly through centroid (x_bar, y_bar), NOT (0, 0)!

WATCH / Regressing x on y gives slope b_xy = S_xy / S_yy != 1 / b_yx: their product b_yx * b_xy = r^2 <= 1 equals squared Pearson correlation!

Standard errors of slope sigma_b & intercept sigma_a (OL2)

Notice b = sum_i c_i y_i is linear in y_i with weights c_i = (x_i - x_bar)/S_xx satisfying sum c_i^2 = 1/S_xx. Propagating variance gives:

$$\sigma_y = \sqrt{\frac{\sum r_i^2}{n-2}}, \qquad \sigma_b = \frac{\sigma_y}{\sqrt{S_{xx}}}, \qquad \sigma_a = \sigma_y \sqrt{\frac{1}{n} + \frac{\bar{x}^2}{S_{xx}}}$$

WHY / Look at n - 2 in residual scatter sigma_y: fitting TWO parameters (a, b) consumes 2 degrees of freedom (with just n = 2 points, line hits both exactly so residuals = 0)!

Worked full Olympiad exam drill (Paper P30)

Data {(1, 2.1), (2, 3.9), (3, 6.2), (4, 7.8)} with known sigma_y = 0.2: here x_bar = 2.5, y_bar = 5.0, dx = {-1.5, -0.5, +0.5, +1.5}:

$$S_{xx} = 5.0, \;\; S_{xy} = 9.70 \Rightarrow b = 1.94, \qquad a = 0.15$$

Uncertainties: sigma_b = 0.2 / sqrt(5) = 0.089, sigma_a = 0.2 * sqrt(0.25 + 6.25/5.0) = 0.2 * sqrt(1.50) = 0.245 -> y = (0.15 +- 0.25) + (1.94 +- 0.09) x!

WHY / Design rule: placing half your trials near x_min and half near x_max maximizes S_xx (after verifying linearity with interior points)!

Through-origin slope & unequal error bars (WLS)

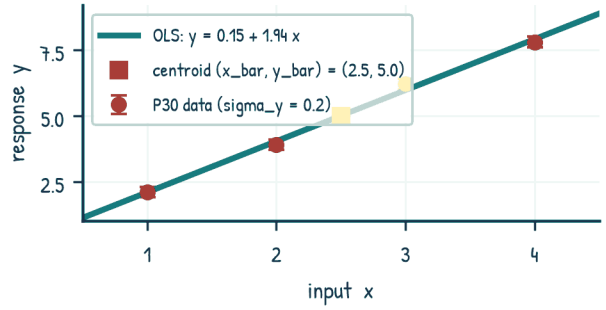
1. If physics GUARANTEES zero intercept (y = b x with zero offset verified), minimising sum (y_i - b x_i)^2 gives:

$$b_{\text{origin}} = \frac{\sum x_i y_i}{\sum x_i^2}, \qquad \sigma_{b,\,\text{origin}} = \frac{\sigma_y}{\sqrt{\sum x_i^2}}$$

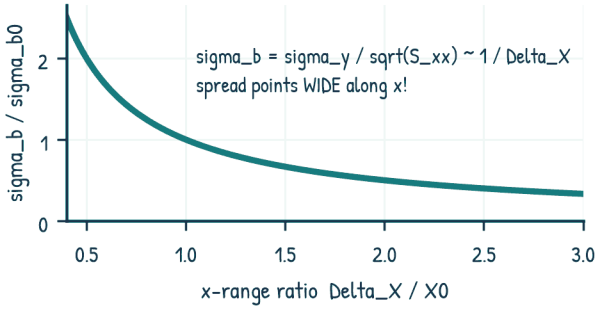
2. When error bars sigma_i vary from point to point (e.g. after taking ln y!), weight every point by inverse variance w_i = 1 / sigma_i^2:

$$w_i = \sigma_i^{-2} \Rightarrow b = \frac{\sum w_i (x_i - \bar{x}_w)(y_i - \bar{y}_w)}{\sum w_i (x_i - \bar{x}_w)^2}$$

Paper P30 fit: OLS line passes strictly through centroid (2.5, 5.0) with b = 1.94 +- 0.09



Design lever for straight-line slopes: doubling x-span halves slope uncertainty sigma_b



REMEMBER / OLS line ALWAYS passes through centroid (x_bar, y_bar) with b = S_xy / S_xx and sigma_b = sigma_y / sqrt(S_xx) — spread x_i wide!

Chi-square goodness of fit, weighted mean & covariance

Testing models quantitatively with chi^2/(n - p), optimal inverse-variance data fusion, and full correlated error propagation.

Chi-square test statistic & degrees of freedom (OL3)

For n independent Gaussian measurements y_i +- sigma_i fitted by a theoretical curve f(x_i; theta_1, ..., theta_p) having p adjustable parameters:

$$\chi^2 = \sum_{i=1}^{n} \left(\frac{y_i - f(x_i)}{\sigma_i} \right)^2, \qquad \nu = n - p, \qquad \chi^2_\nu \equiv \frac{\chi^2}{\nu}$$

WHY / Each standardised residual z_i = r_i / sigma_i ~ N(0, 1) contributes E[z_i^2] = 1, minus p fitted constraints: E[chi^2] = nu and SD(chi^2) = sqrt(2 nu)!

Check P30(c): residuals {+0.01, -0.13, +0.23, -0.11} with sigma = 0.2 give chi^2 = 2.05 for nu = 4 - 2 = 2 (chi^2_nu = 1.03 ~ 1 -> textbook fit!).

Three diagnostic outcomes of reduced chi^2_nu

1. chi^2_nu ~ 1 (within 1 +- sqrt(2/nu)): model AND stated uncertainties sigma_i are mutually consistent.

2. chi^2_nu >> 1 (e.g. chi^2 = 45 for nu = 8): EITHER model is missing real physics (curvature / systematic drift) OR error bars sigma_i were underestimated!

WATCH / 3. chi^2_nu << 1 (e.g. chi^2 = 0.4 for nu = 10): NOT "miraculously accurate data" — your error bars sigma_i are way too conservative (or data was overfitted/fudged)!

Optimal combination of independent labs (weighted mean)

Two or more independent experiments measure the SAME physical constant with different precisions x_1 +- sigma_1, ..., x_k +- sigma_k. Minimising chi^2(mu) = sum (x_i - mu)^2 / sigma_i^2 gives:

$$w_i = \frac{1}{\sigma_i^2}, \qquad \bar{x}_w = \frac{\sum w_i x_i}{\sum w_i}, \qquad \sigma_{\bar{x}_w} = \frac{1}{\sqrt{\sum w_i}} < \min_i \sigma_i$$

WHY / Combined precision ALWAYS beats every individual experiment! If sigma_1 = 0.01 and sigma_2 = 0.10, Run 1 carries weight 100x larger than Run 2 — NEVER take unweighted mean (x1+x2)/2!

Correlated errors & covariance matrix formula

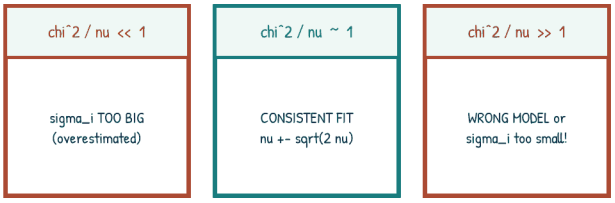
Whenever two fitted quantities (such as OLS intercept a and slope b, with Cov(a, b) = -x_bar sigma_y^2 / S_xx < 0!) derive from the SAME dataset, use covariance C_xy = <dx dy>:

$$\sigma_f^2 = \left(\frac{\partial f}{\partial x}\right)^2 \sigma_x^2 + \left(\frac{\partial f}{\partial y}\right)^2 \sigma_y^2 + 2\left(\frac{\partial f}{\partial x}\right)\left(\frac{\partial f}{\partial y}\right) \mathrm{Cov}(x, y)$$

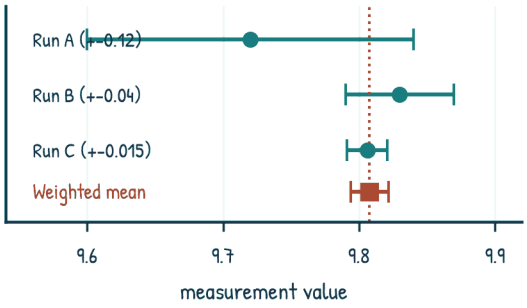
WHY / Look how nicely this explains extrapolating y_hat(x) = a + b x: plug Cov(a, b) in and cross-term cancels x_bar to give exact confidence fan:

$$\sigma_{\hat{y}(x)} = \sigma_y \sqrt{\frac{1}{n} + \frac{(x - \bar{x})^2}{S_{xx}}} \quad (\text{tightest right at centroid } x = \bar{x}!)$$

Diagnostic meaning of reduced chi-square chi^2_nu = chi^2 / (n - p)



Inverse-variance weighting w_i = 1/sigma_i^2 pulls combined estimate to sharpest run



REMEMBER / A trustworthy fit of p parameters to n data points satisfies chi^2 ~ (n - p) +- sqrt(2(n - p)); combining measurements weights strictly by 1/sigma_i^2!

Poisson counting statistics, background & dead time

Exact sqrt(N) shot noise, rigorous background-subtraction formula (fixing P35c), optimal time splitting, and Geiger dead time.

Poisson event counts & the universal sqrt(N) law (OL6)

Independent random point events at mean rate R over time t follow Poisson law P(k) = mu^k exp(-mu) / k! with mean mu = R t and exact standard deviation:

$$\sigma_N = \sqrt{N}, \qquad \frac{\sigma_N}{N} = \frac{1}{\sqrt{N}}, \qquad R = \frac{N}{t} \pm \frac{\sqrt{N}}{t} = R \pm \sqrt{\frac{R}{t}}$$

Worked OL6: N = 10,000 counts in 600 s gives 1% precision (16.67 +- 0.17 s^-1). Reaching 0.1% demands N = 10^6 counts -> 100x longer run (16.7 hours)!

WATCH / CRITICAL TRAP: apply sqrt(.) ONLY to raw integer COUNTS N, NEVER to rate R! If 100 counts arrive in 10 s, R = 10 +- 1 s^-1 (NOT 10 +- sqrt(10) = 10 +- 3.16)!

Exact background subtraction (audit of Paper P35c!)

Count source+background for t_on (observing N_tot = R_tot t_on) and pure background separately for t_off (observing N_b = R_b t_off):

$$R_{\text{net}} = \frac{N_{\text{tot}}}{t_{\text{on}}} - \frac{N_b}{t_{\text{off}}}, \qquad \sigma(R_{\text{net}}) = \sqrt{\frac{R_{\text{tot}}}{t_{\text{on}}} + \frac{R_b}{t_{\text{off}}}} = \sqrt{\frac{R_s + R_b}{t_{\text{on}}} + \frac{R_b}{t_{\text{off}}}}$$

WHY / Audit note on P35(c): with R_tot = 250/s (R_s = 200/s) and R_b = 50/s, if R_b is known with zero error then sigma(R_net) = sqrt(2500)/10 = 5.0/s; ONLY when R_b is ALSO measured for 10 s (N_b = 500) does variance become (2500 + 500)/10^2 -> 5.48/s!

Optimal beam-time split for fixed total lab time T

Given fixed total detector time T = t_on + t_off, minimise variance V = R_tot / t_on + R_b / (T - t_on) with respect to t_on:

$$-\frac{R_{\text{tot}}}{t_{\text{on}}^2} + \frac{R_b}{t_{\text{off}}^2} = 0 \Rightarrow \frac{t_{\text{on}}}{t_{\text{off}}} = \sqrt{\frac{R_{\text{tot}}}{R_b}}$$

WHY / Clean physical limits: if signal >> background (R_tot >> R_b), spend almost all time on source; if weak signal sits atop huge background (R_tot ~ R_b), split 50% / 50%!

Detector dead time tau (non-paralysable model)

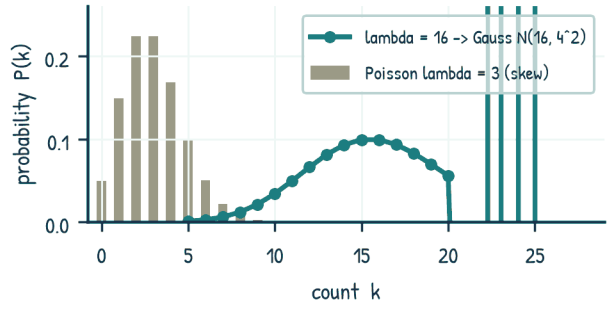
After each registered click, a Geiger/scintillation counter stays blind for dead time tau. In 1 second it registers m clicks and is dead for total fraction m * tau:

$$m = R_{\text{true}}(1 - m\tau) \Rightarrow R_{\text{true}} = \frac{m}{1 - m\tau}, \qquad m = \frac{R_{\text{true}}}{1 + R_{\text{true}}\tau}$$

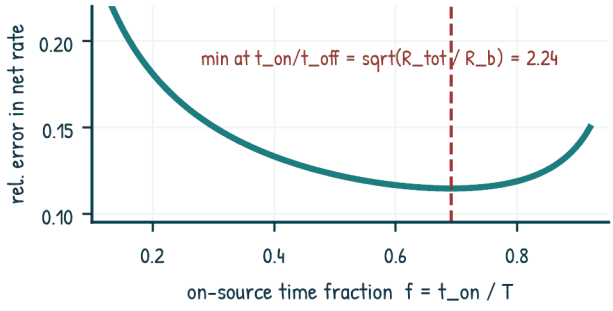
WHY / At intense rates R_true >> 1/tau, observed count rate saturates at ceiling 1/tau AND sub-Poisson anti-bunching breaks naive sqrt(N) statistics!

Two-source lab calibration: measure m1, m2 separately and m12 together -> m1 + m2 > m12 reveals dead time tau directly without knowing R_true!

Poisson counts converge to Gaussian mean N and standard deviation sqrt(N) for N >> 10



Optimal beam-time allocation between gross rate R_tot = 250/s and background R_b = 50/s



REMEMBER / Subtracting background REQUIRES ADDING variances: sigma(R_net) = sqrt(R_tot / t_on + R_b / t_off); NEVER substitute R_net inside the square root!

Olympiad experiment design: 0.1% g-budget & natural units

End-to-end error budget engineering (OL5 / P32), systematic corrections on real pendulums, and symmetry scaling laws.

Engineering a 0.10% measurement of g (OL5, P32)

Available gear: ruler (dL = 1 mm), stopwatch (dt = 0.01 s), string up to 2 m, steel bob. Target: dg / g < 0.10% from g = 4 pi^2 L / T^2:

$$\frac{\Delta g}{g} = \frac{\Delta L}{L} + 2\frac{\Delta t_{\text{sw}}}{NT} + 2\left(\frac{\theta_0^2}{16}\right)$$

Design choices: 1. Max out length L = 2.000 m so dL/L = 1/2000 = 0.050%. 2. Now T = 2.84 s; time N = 50 continuous swings (142 s) so 2 dt/(N T) = 0.014%!

3. Restrict launch angle to theta_0 = 2 deg = 0.0349 rad so 2(theta_0^2/16) = 0.015%! Total budget = 0.050 + 0.014 + 0.015 = 0.079% < 0.10%!

Three hidden physical systematics in real pendulums

To trust 0.1% accuracy (not merely precision!), account for three real-world physics corrections beyond point-mass vacuum theory:

$$I_{\text{pivot}} = mL^2 + \tfrac{2}{5}mr_{\text{bob}}^2 \Rightarrow L_{\text{eff}} = L_{\text{cm}}\left(1 + \frac{2r_{\text{bob}}^2}{5L_{\text{cm}}^2}\right)$$

$$g_{\text{eff}} = g\left(1 - \frac{\rho_{\text{air}}}{\rho_{\text{bob}}}\right) \quad (0.015\%\ \text{steel},\ 0.05\%\ \text{glass})$$

WHY / Plus time zero-crossings AT EQUILIBRIUM (fastest bob speed -> sharpest visual trigger!), NEVER at extreme turning points where v = 0!

Relativistic corrections via symmetry + dimensions (OL10, P36)

How must momentum p = m v f(v/c) or rod length L = L0 g(v/c) behave at small speeds beta = v/c BEFORE invoking full Lorentz algebra?

$$\mathbf{p} = m\mathbf{v}\, f\left(\frac{\mathbf{v}\cdot\mathbf{v}}{c^2}\right) = m\mathbf{v}\left(1 + \alpha_1\frac{v^2}{c^2} + O(\beta^4)\right)$$

WHY / Spatial isotropy + smoothness at v = 0 forbids odd linear terms +-v/c in scalar factor f! Special relativity fixes alpha_1 = +1/2 for momentum (1% shift at v = 0.141 c, P36) and -1/2 for Lorentz contraction!

Natural units c = hbar = k_B = 1 (theoretical Olympiad tool)

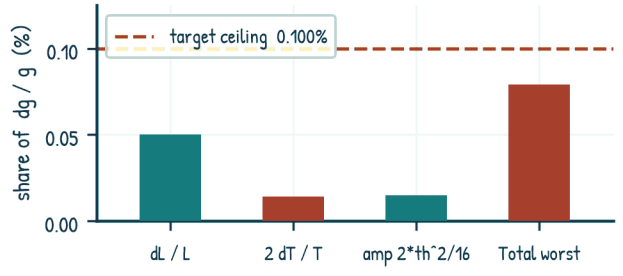
In particle/astrophysics Olympiad problems using natural units (hbar = c = k_B = 1), EVERY mechanical/thermal quantity collapses onto powers of ENERGY [E]:

$$[M] = [T_{\text{temp}}] = [p] = [E], \quad [L] = [t] = [E^{-1}], \quad [A] = [E^{-2}]$$

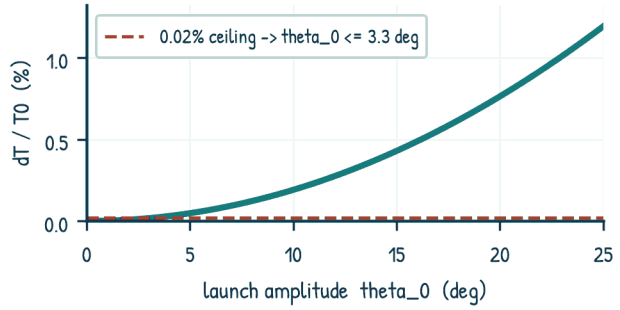
WHY / To restore hbar and c at the end of any derivation, uniquely match [M, L, T]: e.g. energy density [u] = [E^4] restores to u ~ (k_B T)^4 / (hbar c)^3 (Stefan-Boltzmann T^4 scaling in ONE second)!

Handy anchor: hbar c = 197.3 MeV fm = 197.3 eV nm!

OL5 experimental budget (L = 2.000 m, N = 50 swings, theta_0 = 2 deg): 0.079% < 0.1%!



Anharmonic pendulum period drift +theta_0^2 / 16 (theta_0 MUST be in radians!)



REMEMBER / In experiment design, list EVERY random AND systematic term in dg/g BEFORE touching equipment, and engineer every term well below ceiling!